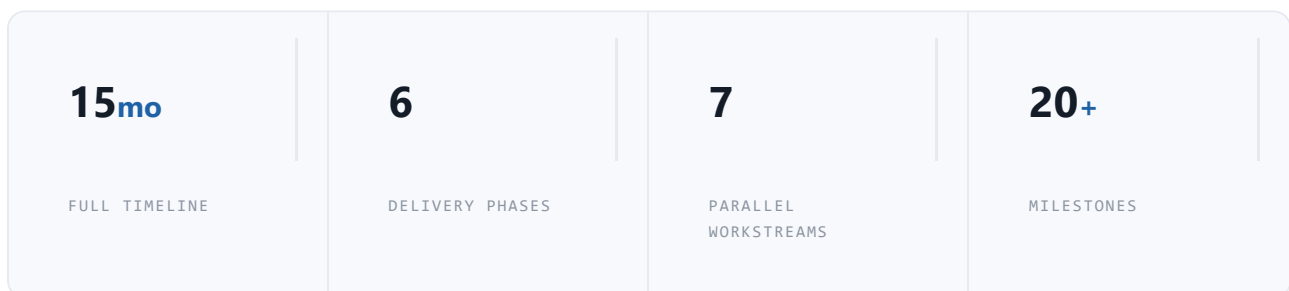


SCALE-UP PROGRAM • TIMELINE EDITION

What gets built, and **exactly when.**

A month-by-month technical delivery timeline for evolving NOVA Terminal into a multi-tenant, multi-broker, intelligent trading platform — every workflow, milestone and deliverable, mapped to the point in time it happens.



A timeline you can hold people to.

This document answers one question in detail: **at each point in the 15-month program, what is being built, and what is delivered?** It maps every technical workstream onto the calendar so progress is measurable, sequenced, and verifiable.

STRUCTURE

Part 1 — Overview

Executive summary, the master 15-month timeline, phase summary, and the seven parallel workstreams that run through it.

STRUCTURE

Part 2 — The timeline, in detail

Each phase broken to the month, with the technical work, deliverables, dependencies and the checkpoint that closes it.

STRUCTURE

Part 3 — Testing & launch

The 7-month real-world testing timeline, go-live sequence, and the post-launch roadmap.

STRUCTURE

Part 4 — Cross-cutting

Technology stack, KPIs & checkpoints, the risk register with timeline dependencies, and the master milestone table.

READING THE MARKERS

A **PHASE** tag marks a delivery phase; a **TRACK** tag marks a workstream. Month labels are targets carrying a built-in comfort buffer of up to ~3 months. Every phase closes on a **demonstrable checkpoint** — something you can see working.

FOUNDATION

The hardest component — a live, reliable brokerage engine — is already built and running in production. This timeline scales proven technology; it does not gamble on discovery.

Eight months to build. Seven to prove. One clear path.

The program runs 15 months end to end: **8 months of engineering** across four build phases, followed by **7 months of real-world and pre-user testing** across two testing phases — closing on a launch-ready, multi-tenant, multi-broker platform.

M1–M8 BUILD (4 PHASES)	M9–M15 TESTING (2 PHASES)	6+ BROKERS AT LAUNCH	99.9% UPTIME TARGET
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THE BUILD (MONTHS 1–8)

P0 Foundation · **P1** Multi-tenancy + broker abstraction · **P2** Subscriptions + onboarding · **P3** Scale, data, AI, mobile & hardening. Closes feature-complete and load-tested.

THE PROVING (MONTHS 9–15)

P4 Closed beta with real traders on real brokers · **P5** Scale/soak testing, security pen-test, DR drills and compliance review. Closes launch-ready and signed off.

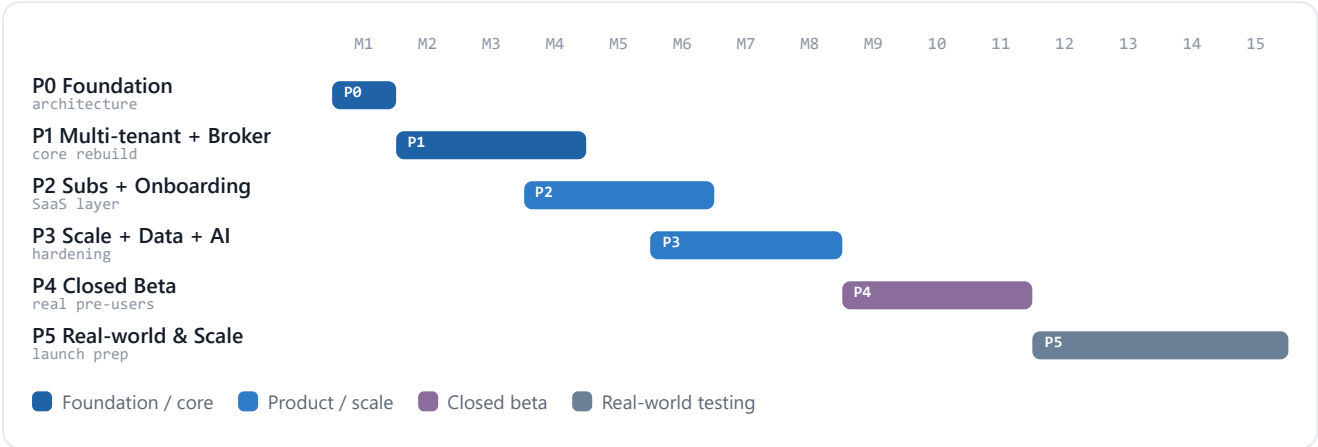
THE PRINCIPLE

Every phase ends in something demonstrable.

No phase closes on a status report. Each closes on a checkpoint you can see and test — from "two tenants trading on two brokers simultaneously" to "a stranger signs up, connects a broker, and places a live order unaided."

Everything, on one calendar.

Six phases and seven workstreams laid across 15 months. Workstreams overlap deliberately — the platform, broker, data, AI, infrastructure, mobile and QA tracks advance in parallel, synchronized at phase checkpoints.



READING THE CALENDAR

Bars show when each phase is the primary focus; the workstreams (next page) run continuously beneath them. Month labels are targets with a comfort buffer of up to ~3 months.

END M8 **Build complete** — feature-complete, load-tested platform enters the testing program.

END M15 **Launch-ready** — scaled, secured, audited and signed off for public onboarding.

Parallel tracks, one synchronized program.

Work is organized into seven tracks that run in parallel and converge at each phase checkpoint. This is why an 8-month build is achievable: the tracks progress simultaneously, not sequentially.

TRACK 1

Platform & Multi-tenancy

Tenant model, isolation, RLS, entitlements, the core service backbone. Heaviest in M1–M4.

TRACK 2

Broker Integrations

Broker abstraction layer, native adapters (IBKR, Schwab, Alpaca), aggregator. M2–M6.

TRACK 3

Market Data

Vendor-grade real-time feed, streaming fan-out, session-aware pricing. M5–M8.

TRACK 4

Intelligence & AI

Alerts engine at scale, co-pilot, sentiment, analytics. M6–M8, refined through beta.

TRACK 5

Infrastructure & SRE

Cloud, containers, autoscale, observability, CI/CD, DR. Foundational M1, hardened M6–M8.

TRACK 6

Apps & Experience

Web app, onboarding, billing UX, mobile apps, admin console. M4–M8.

TRACK 7

Quality, Security & Compliance

Test automation, sandbox conformance, security, SOC 2 track, pen-test, DR drills. Continuous; intensive M9–M15.

SYNC

Phase checkpoints

All tracks converge on a demonstrable milestone at the end of each phase — the synchronization points of the program.

The timeline, phase by phase.

Each build phase, broken to the month: the technical work in flight, what is delivered, the dependencies, and the demonstrable checkpoint that closes it.

PHASE	WINDOW	FOCUS	CLOSING CHECKPOINT
P0	Month 1	Foundation & architecture	Skeleton on staging
P1	Months 2–4	Multi-tenancy + broker abstraction	2 tenants · 2 brokers · live
P2	Months 4–6	Subscriptions + onboarding + breadth	Self-serve signup → live order
P3	Months 6–8	Scale, data, AI, mobile & hardening	Feature-complete, load-tested
P4	Months 9–12	Closed beta — real pre-users	Cohort exit criteria met
P5	Months 13–15	Real-world, scale & compliance	Launch-ready, signed off

Before a line of feature code: lock the decisions that everything else depends on. One month, entirely about getting the foundations provably right.

WHAT HAPPENS, WEEK BY WEEK

MONTH 1 • WEEK 1

Tenant & data model

Design the multi-tenant schema and row-level-security model; decide isolation boundaries and per-tenant configuration.

MONTH 1 • WEEK 2

Broker Adapter contract

Specify the universal broker interface (authenticate, quote, place, modify, cancel, verify, positions, executions) that every broker will implement.

MONTH 1 • WEEK 3

Security & threat model

Credential vault design (OAuth, zero-password), encryption/KMS, tenant-isolation threat model; initiate native broker API approvals (Schwab, IBKR).

MONTH 1 • WEEK 4

Cloud & CI/CD scaffold

Containerized service skeleton, staging environment, pipelines, infrastructure-as-code, observability baseline.

CHECKPOINT

Approved architecture + running skeleton on staging, and broker API approval processes started (a long-lead item begun early on purpose).

WHY A WHOLE MONTH ON FOUNDATIONS

Multi-tenancy, the broker contract and the security model are decisions that are expensive to change later. Getting them right in Month 1 is what makes Months 2–8 fast.

The core rebuild. Re-platform the proven engine behind the abstraction, make it multi-tenant, and light up the first new brokers.

MONTH BY MONTH

MONTH 2

Multi-tenancy foundation goes live

Tenant identity on every request; row-level security enforced end-to-end; per-tenant config, entitlements and rate limits at the gateway. The proven IBKR engine is moved behind the new tenant boundary.

MONTH 3

Broker abstraction + IBKR on the new layer

The Broker Adapter interface is implemented; the existing IBKR integration is re-platformed onto it — same proven reliability, now broker-agnostic upstream.

MONTH 4

Schwab + Alpaca adapters + session pool

First new native adapters (OAuth); the self-healing session becomes a managed, per-tenant pool that keeps each broker link alive independently.

CHECKPOINT

Two independent tenants trading on two different brokers, simultaneously — the proof that the abstraction and isolation both work.

Multi-tenancy, built to isolate by construction.

Delivered across Months 2–4. Isolation is structural — a tenant cannot reach another's data even if application code has a bug.

MONTH 2

Identity on every request

Verified tenant identity attached at the edge; the gateway rejects any unscoped request before it reaches a service.

MONTH 2

Row-level security

The database enforces tenant boundaries per row — queries are physically unable to return another tenant's data.

MONTH 3–4

Per-tenant entitlements

Feature flags, plan limits and rate limits are tenant-scoped and enforced at the gateway — the backbone the subscription tiers later plug into.

MONTH 4

Per-tenant session pool

The production session auto-revive, generalized: each tenant's broker link is kept alive, health-checked and revived independently, scheduled across worker fleets and autoscaled.

REUSES PROVEN WORK

The self-healing session logic already runs in production for one account. P1 generalizes it to thousands — scaling a solved problem, not solving a new one.

The Broker Abstraction Layer.

One interface every broker implements; the engine above never knows which broker it is talking to. First brokers land in Months 3–4; breadth follows in P2.

THE ADAPTER CONTRACT (DEFINED P0, IMPLEMENTED FROM M3)

CAPABILITY	WHAT THE ADAPTER PROVIDES	FIRST LIVE
Authenticate	OAuth connect, token refresh, session health, revoke	M3 (IBKR)
Quote	Real-time snapshot & streaming	M3
Place / modify / cancel	All order types, brackets, extended hours, confirmation loop	M3
Verify	Confirm the order actually reached the broker post-submit	M3
Positions & balances	Holdings, average cost, buying power, margin	M3
Executions	Trade history for archive & P&L	M3

M3–M4

Native adapters

IBKR (re-platformed), then Schwab & Alpaca via OAuth — full capability, lowest latency.

M5–M6 (P2)

Aggregator adapter

One integration (SnapTrade / Plaid-class) unlocks many brokers at once — breadth without dozens of separate builds.

TIMELINE DEPENDENCY

Native broker programs need API approval (started in P0). The aggregator path runs in parallel so broker breadth is never blocked on a single approval queue.

Turn the engine into a product people can buy and self-serve — and widen broker coverage through the aggregator.

MONTH BY MONTH



MONTH 4-5

Self-serve onboarding

Sign up → connect broker (OAuth) → fund-check → trade, unaided. Identity, email verification, and the "connect your broker" flow go live.



MONTH 5

Billing & entitlements

Stripe-class billing, plans (Starter/Pro/Elite), monthly & annual cycles, trials, proration; entitlements wired to the per-tenant flags built in P1.



MONTH 5-6

Aggregator brokers online

Many additional brokers unlocked via the aggregation layer; broker picker expands.



MONTH 6

Statements & notifications service

Automated daily/monthly statements; instant trade & alert notifications; multi-channel delivery groundwork.

CHECKPOINT

A stranger can sign up, subscribe, connect a broker, and place a live order — unaided.

Onboarding & billing — the SaaS layer.

M4-5 • ONBOARDING

Connect-your-broker flow

OAuth authorization on the broker's own page; NOVA stores only a scoped, revocable token in the encrypted vault — never a password.

M5 • BILLING

Plans & lifecycle

Monthly/annual, trials, coupons, upgrades with proration, dunning & recovery, invoices — a complete subscription spine.

M5 • ENTITLEMENTS

Feature gating

Central entitlements gate broker count, alert volume, F&O, real-time data, AI and automation per tier — enforced in real time.

M6 • COMMS

Statements & alerts

Contract-note statements auto-emailed; instant trade/alert notifications; SMS/WhatsApp/Telegram channels scaffolded.

ONBOARDING SEQUENCE (TARGET: UNDER 5 MINUTES)

STEP	ACTION	WHAT THE PLATFORM DOES
1	Sign up	Create tenant, verify email
2	Choose plan	Start trial / subscribe; entitlements set
3	Connect broker	OAuth on broker page → scoped token to vault
4	Sync account	Positions, balances, buying power load
5	Trade	Alerts + live order desk active

Make it fast, intelligent and resilient — and take it to feature-complete, load-tested, ready for the testing program.

MONTH BY MONTH

MONTH 6-7

Shared market-data feed

Vendor-grade real-time feed ingested once and streamed to all tenants; session-aware pricing; the price-resilience already proven in production, designed in from day one.

MONTH 7

Scale infrastructure + observability

Autoscaling per service, distributed tracing, trading-specific SLOs (order-success, session uptime, data-staleness, latency), paging & runbooks.

MONTH 7

AI co-pilot + intelligence

Grounded conversational co-pilot over the trader's live account; sentiment; alerts engine scaled as a service; risk guardrails & trade journal.

MONTH 7-8

Mobile apps to store-ready

iOS & Android with full trading, alerts, push, biometrics, widgets.

MONTH 8

Security & performance hardening

Load tests, security review, dependency/secret scanning, pen-test preparation.

CHECKPOINT

Feature-complete, load-tested platform — end of the 8-month build; enters the testing program.

Data, intelligence & scale.

M6-7 • MARKET DATA

Ingest-once, fan-out

One subscription per symbol streamed over WebSockets to all interested tenants; entitlement-tiered real-time vs delayed; cost amortized across the base.

M7 • AI

Grounded co-pilot

Retrieval-grounded over live positions/orders/market data, with sources — explains, never fabricates; "not advice" guardrails.

M7 • SAFETY

Risk guardrails

Per-trade risk sizing, daily-loss limits, over-concentration & earnings-hold warnings, kill-switch — discipline enforced by software.

M7 • INFRA

Autoscale + observability

Horizontal scaling per service, tracing, trading SLOs, DR runbooks, blue-green releases.

99.9%

UPTIME TARGET

<1_s

QUOTE LATENCY P95

<15_m

RECOVERY OBJECTIVE

Multi

AVAILABILITY ZONES

END OF BUILD

At the close of Month 8 the platform is feature-complete and load-tested. Nothing ships to real traders until it passes the 7-month testing program that follows.

Seven months earning the right to launch.

In live trading, "probably works" is a defect. The build is followed by seven months of real-world and pre-user testing before the platform ever asks a stranger to connect their brokerage.

P4 CLOSED BETA • M9-12	P5 SCALE & COMPLIANCE • M13-15	7mo TOTAL TESTING	M15 LAUNCH-READY
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A hand-picked group of real traders on real broker connections, closely instrumented. This is where the platform meets reality.

MONTH BY MONTH



MONTH 9

Invite-only onboarding

First cohort onboarded on real brokers with real (small) money; heavy instrumentation on every session, order and data path.



MONTH 10

Feedback & rapid iteration

Weekly feedback loop; fix, tune, and re-ship; refine alerts honesty, onboarding friction, and reliability edge cases.



MONTH 11

Reliability & support playbooks

Support workflows, incident runbooks, and reliability tuning under real usage; widen the cohort.



MONTH 12

Exit-criteria review

Measure against hard bars — order-success, session uptime, data accuracy, retention — before advancing.

CHECKPOINT

Sustained clean trading across the cohort; defined exit criteria met.

WHY REAL MONEY, REAL BROKERS

A sandbox cannot surface the failure modes that only appear with live sessions, real fills, and real market conditions. The beta is designed to find them while the stakes are contained.

Widen the cohort and push the platform to its limits — performance, security, disaster recovery, and compliance — before the doors open.

MONTH BY MONTH



MONTH 13

Expanded beta + load / soak

Larger cohort; sustained high-concurrency load tests and multi-day soak runs; chaos/failure injection (broker outages, session drops) to prove graceful degradation.



MONTH 14

Security pen-test + DR drills

Independent third-party penetration test; disaster-recovery and failover drills against the runbooks.



MONTH 15

Compliance & SOC 2 readiness

Formal compliance review, SOC 2 readiness assessment, final hardening and sign-off.

CHECKPOINT

Launch-ready platform — scaled, secured, audited and signed off for public onboarding.

Open the doors — carefully.

Launch is staged, not flipped. Intake is capped and lifted only as the reliability objectives (SLOs) continue to hold under real load.

LAUNCH • WEEKS 1-2

Soft launch

Waitlist & referrals; capped daily intake; every new tenant monitored closely; SLOs watched in real time.

LAUNCH • WEEKS 3-4

Ramp

Intake caps lifted step by step as order-success, uptime and latency hold; support scaled to match.

LAUNCH • MONTH 2

Public launch

Open signup, marketing switched on, full self-serve at scale.

LAUNCH • ONGOING

Scale

Grow brokers, regions and features on a proven, monitored platform.

DISCIPLINE

Caps exist so that reliability leads growth, never the other way around — a trading platform earns trust by never breaking under its own success.

The platform keeps compounding.

Launch is the start line. The funded next horizon deepens breadth, depth and ecosystem on an already-proven base.

BREADTH More brokers & regions • Additional native brokers • New regions & currencies • Localized languages	DEPTH Advanced product • Copy / social trading • Full automation studio • Advanced options tooling	ECOSYSTEM Open & enterprise • Public API GA • Partner integrations • White-label / institutional
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EXPANDED CAPABILITIES (SEQUENCED ACROSS POST-LAUNCH)

Intelligence Deeper co-pilot, pattern & unusual-activity detection, portfolio AI review, predictive risk alerts, news/sentiment expansion.	Trading tools Backtesting studio (full), strategy builder, automated execution, options strategy builder, basket & rebalancing.
Analytics Advanced screener, deeper journal analytics, heatmaps, tax-ready reports.	Engagement Leaderboards, education hub, watchlist sharing, mobile widgets/wearables, multi-channel alerts.

— PART 4

The technical detail, and how it all holds together.

The architecture and each major subsystem, mapped to when it is built — plus the cross-cutting concerns (security, data, reliability), the stack, KPIs, dependencies, risk, team and the master deliverables table.

Layered, isolated, broker-agnostic.

Every tenant is logically isolated; the engine talks to an abstraction, never a specific broker; each layer scales on its own. Established in P0–P1, hardened through P3.

LAYER	COMPONENTS	BUILT
Client	Web app (React/TanStack), mobile apps, admin console	P2–P3
Edge & identity	API gateway (rate-limit, WAF), auth/SSO, billing, tenant router, public API	P1–P2
Core services	Broker abstraction, order engine, alerts engine, AI/co-pilot, analytics/P&L	P1–P3
Data & integration	Market data, credential vault, tenant DB (RLS), cache/queue, notifications	P0–P3

Stateless services

Core services scale horizontally behind the gateway; state lives in the data tier.

Event-driven

Fills, alerts and statements flow through a queue — resilient, replayable, decoupled.

Failure-isolated

A broker or vendor outage degrades one capability, never the whole platform.

The order engine — correctness is non-negotiable.

Where a bug costs real money. Built to make silent failure impossible and to carry every professional order type. This is proven in production today and re-platformed onto the abstraction.

Confirmation loop

Broker order questions (warnings, price caps) answered programmatically so nothing is dropped mid-submit.

Post-submit verification

After every submit, poll the real order book and alert loudly if the order never appeared — the dead-bridge silent drop, caught.

Full order types

Market, Limit, Stop, MIT (touch→market), bracket/OCA, trailing stop; GTC by default; extended-hours execution.

Pre-trade safety

Risk-based sizing, over-concentration & daily-loss guards, buying-power checks, bracket sanity, kill-switch.

ALREADY SHIPPED

Post-submit verification, GTC-by-default and MIT are live in production today. P1 moves them behind the broker abstraction so they work identically across every broker.

Market data — one fresh price, fanned out to everyone.

The free feed used today freezes and lags. At scale, NOVA runs a shared, vendor-grade real-time feed — ingested once, streamed to every tenant.

Vendor-grade ingest

A professional real-time vendor feeds a single ingestion service — no per-user scraping.

Fan-out streaming

One subscription per symbol, streamed over WebSockets to all interested tenants — efficient at scale.

Session-aware truth

Broker real-time in regular hours; consolidated tape pre/after-market — the displayed price always matches reality.

LESSON ALREADY BANKED

Production already persists the last real price so a feed hiccup never reverts a card to stale data. That hard-won resilience is designed into the shared feed from day one.

Intelligence — explainable, grounded, safe.

NOVA's intelligence is grounded in the trader's real account and real market data — it explains, it does not hallucinate advice. Guardrails keep it honest and compliant.

Grounded co-pilot

Answers over the trader's live positions, orders and market data — retrieval-grounded, with sources, never fabricated numbers.

Signal & scoring services

The alert engine's scoring and honest quality gates run as scalable services; models/rules versioned and backtested before shipping.

Sentiment & NLP

News and filings scored per symbol feed alerts and the co-pilot's context.

Guardrails

Clear "not advice" framing, honest uncertainty, refusal to over-promise — integrity as a feature and a compliance safeguard.

Security & compliance, across the timeline.

Security is a first-class deliverable engineered from Day 1 — not a phase-end checkbox — because NOVA holds other people's brokerage links.

P0 · MONTH 1

Threat model & vault design

Zero-password OAuth architecture, encrypted credential vault, KMS, per-tenant key separation designed up front.

P1-P2

Structural isolation

Row-level security, scoped tokens, gateway-enforced tenancy — cross-tenant access impossible by construction.

P3 · MONTH 8

Hardening & scanning

Dependency & secret scanning, audit logging, anomaly detection on auth/trading; pen-test preparation.

P5 · MONTH 14

Third-party penetration test

Independent security assessment; remediation before launch.

P5 · MONTH 15

SOC 2 readiness & compliance review

Formal SOC 2 Type II readiness; GDPR/privacy review; broker-ToS conformance sign-off.

CLEAN REGULATORY POSTURE

Execution & custody stay with the trader's regulated broker — NOVA never takes custody of funds, avoiding the biggest regulatory burden while delivering the full experience.

The record, owned; the platform, observable.

DATA ARCHITECTURE (P0 DESIGN, P1-P3 BUILD)

Tenant data store

Managed PostgreSQL with row-level security, encryption at rest, point-in-time recovery.

All-time archive

Every execution archived independently of the broker's short retention window — proven in production today.

Event stream

Fills, alerts, statements as replayable events for analytics and audit.

OBSERVABILITY & SRE (P3 HARDENING, CONTINUOUS THEREAFTER)

Trading-specific SLOs

Order-success rate, session uptime, data-staleness, quote latency — tracked as first-class objectives with paging on breach.

Self-healing

The production session auto-revive, generalized to the fleet with health checks and automatic recovery; DR runbooks and drills.

What it is built with.

LAYER	TECHNOLOGY DIRECTION	PRIMARY PHASE
Frontend	React + TanStack (web) · shared design system · native-shell mobile	P2–P3
Backend	Node/TypeScript services · REST + WebSocket · event-driven jobs	P1–P3
Data	PostgreSQL (row-level security) · Redis cache/queue · object storage	P0–P1
Broker	Universal adapter · native (IBKR, Schwab, Alpaca) + aggregation layer	P1–P2
Market data	Vendor-grade real-time feed · shared streaming fan-out	P3
AI	Grounded LLM co-pilot · scoring/sentiment services · versioned models	P3
Platform	Containers + orchestration (Kubernetes-class) · KMS · Stripe-class billing · IaC · observability	P0, P3
Security	OAuth · encrypted vault · TLS 1.2+ · AES-256 · per-tenant keys	P0–P5

BUILD - VS - BUY

Managed services and the broker aggregator are used where they save time without compromising control — protecting the timeline. Native adapters and the core engine are built for depth and reliability.

What "on track" is measured by, at each phase.

PHASE	CHECKPOINT (DEMONSTRABLE)	KEY METRICS GATED
P0	Skeleton on staging; approvals started	Architecture sign-off; CI green
P1	2 tenants · 2 brokers · simultaneously	Isolation verified; order-success in sandbox
P2	Self-serve signup → live order	Onboarding completion; billing correctness
P3	Feature-complete, load-tested	Latency p95 < 1s; autoscale under load
P4	Cohort exit criteria met	Order-success ≥ 99.5%; session uptime; retention
P5	Launch-ready, signed off	0 critical security findings; DR verified; SOC 2 readiness

99.9% UPTIME TARGET	99.5% ORDER-SUCCESS	<1s QUOTE LATENCY P95	<5m SIGNUP → FIRST ORDER
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What must happen before what.

A few dependencies drive the schedule. They are surfaced up front and managed — most notably broker API approvals, which are long-lead and started in P0.

ITEM	DEPENDS ON	TIMELINE HANDLING
Broker abstraction	Adapter contract (P0)	Contract locked M1 so adapters build in parallel from M3
Native broker adapters	Broker API approvals	Approvals started M1 ; aggregator delivers breadth in parallel so no single approval blocks
Subscriptions	Per-tenant entitlements (P1)	Entitlement backbone built M3–4, billing plugs in M5
Shared market data	Vendor selection & terms	Vendor validated in P3 (M6); resilience patterns already proven
Closed beta	Feature-complete build	Beta starts only after the M8 checkpoint passes
Launch	P5 sign-off	Staged go-live only after security + compliance sign-off

THE ONE EXTERNAL LEVER

Native broker approvals are partly outside our control — which is exactly why they start in Month 1 and why the aggregator path runs alongside them. The product is never blocked waiting on a single broker.

Named, and managed on the timeline.

RISK	EXPOSURE	MITIGATION (AND WHEN)
Broker API approval delays	Medium	Approvals started P0; aggregator delivers breadth in parallel; native adapters sequenced
Security / credential breach	High	Zero-password OAuth, vault, isolation from P0; pen-test P5
Per-tenant session scale	Medium	Proven self-healing session generalized to a managed pool in P1; load-tested P3
Market-data cost / licensing	Medium	Shared feed amortized; entitlement-tiered; vendor terms validated P3
Regulatory / broker ToS	Medium	OAuth + broker custody; SOC 2/privacy from P0; review P5
Schedule slip	Low–Med	Built-in ~3-month comfort buffer on all month targets; milestone-gated

The right effort, at the right time.

A small, senior, full-stack team that already knows the codebase — no ramp-up, no hand-off risk. Effort concentrates by workstream as each phase demands.

FUNCTION	PEAK PHASES	FOCUS
Platform & backend	P0–P3	Multi-tenancy, core services, scale
Broker integrations	P1–P2	Abstraction + native + aggregator adapters
Frontend & mobile	P2–P3	Web, onboarding, billing UX, mobile
Data & AI	P3	Market data, co-pilot, analytics
Infra & SRE	P0, P3, P5	Cloud, autoscale, observability, DR
QA & security	Continuous, peak P4–P5	Test automation, sandbox conformance, pen-test
Program & support	All phases	Delivery governance, beta & support ops

GOVERNANCE

Milestone-based delivery with a demonstrable deliverable at each gate, regular written progress reporting, and a shared board tracking scope, risk and schedule against this timeline.

Every deliverable, on the calendar.

PHASE	WINDOW	DELIVERABLES	CHECKPOINT
P0	M1	Architecture, adapter spec, security model, CI/CD scaffold, approvals started	Skeleton on staging
P1	M2–4	Tenant isolation, IBKR on abstraction, Schwab + Alpaca adapters, session pool	2 tenants · 2 brokers
P2	M4–6	Onboarding, billing & entitlements, aggregator brokers, statements/notifications	Self-serve → live order
P3	M6–8	Shared data feed, autoscale + observability, AI co-pilot + guardrails, mobile, hardening	Feature-complete, load-tested
P4	M9–12	Closed beta, instrumentation, feedback loop, support playbooks	Cohort exit criteria met
P5	M13–15	Scale/soak tests, pen-test, DR drills, SOC 2 readiness, sign-off	Launch-ready

TIMELINE NOTE

All month-based dates are targets carrying a built-in comfort buffer of up to ~3 months. Every phase closes on a demonstrable checkpoint — progress proven by what works.

The six pillars, mapped to the calendar.

Every core capability the trader experiences, and the phase in which it lights up.

PILLAR	WHAT IT DOES	LIVE FROM
Universal Broker Connect	Link any supported broker securely (OAuth); one identity, many brokers	P1–P2
AI Stock Alerts	Honest, backtested setups with entry/target/stop and quality gates	P1 (exists), scaled P3
AI F&O / Options	Derivatives with mandatory capital-risk stops; fast option chain	P1 (exists), scaled P3
Intelligent Order Desk	All order types, GTC default, post-submit verification, safety rails	P1 (exists)
Portfolio, P&L & Analytics	Real-time, session-aware, all-time archive, FIFO realized P&L	P1 (exists), enriched P3
Statements & Comms	Automated statements + instant multi-channel notifications	P2

"EXISTS" = ALREADY IN PRODUCTION

Four of the six pillars already run today in the single-tenant platform. The scale-up makes them multi-tenant and multi-broker — it does not re-invent them.

Beyond the pillars — what's in the 15 months, and what's next.

The highest-leverage additions are inside the 15-month scope; the remainder are the funded post-launch roadmap.

IN THE 15-MONTH SCOPE (BY P3)

- NOVA Co-pilot (grounded AI assistant)
- Paper trading / simulator
- Risk guardrails + daily-loss kill-switch
- Trade journal & performance analytics
- News & sentiment (core)
- Multi-channel alerts (push/email + SMS/WhatsApp/Telegram)
- Advanced screener (core)

POST-LAUNCH ROADMAP

- Backtesting studio (full) & strategy builder
- Automated execution (rules engine)
- Options strategy builder (spreads/greeks)
- Copy / social trading & leaderboards
- Public API & webhooks
- Pattern & unusual-activity detection
- Education hub, localization, wearables

SEQUENCING LOGIC

Anything that raises trust, safety and daily engagement (co-pilot, paper trading, guardrails, journal) ships inside the build. Heavier standalone products (automation, copy-trading, API) follow on a proven base.

How the product is packaged.

Entitlements (built P1, wired to billing P2) gate features by tier. Packaging is illustrative of structure.

CAPABILITY	STARTER	PRO	ELITE
Broker links	1	Multiple	Multiple
AI stock alerts	Yes	Yes	Yes
F&O / options intelligence	—	Yes	Yes
Order desk + guardrails	Yes	Yes	Yes
Real-time vendor data	Delayed	Yes	Priority
Paper trading	Yes	Yes	Yes
NOVA Co-pilot	—	Yes	Yes
Trade journal & analytics	Basic	Yes	Yes
Backtesting / automation	—	Limited	Full
Public API & webhooks	—	—	Yes

When each broker comes online.

BROKER	METHOD	CAPABILITY	LIVE FROM
Interactive Brokers	Native (in production)	Full trade + data	Now / P1
Charles Schwab	Native — Trader API (OAuth)	Full trade + data	P1 (M4)
Alpaca	Native — REST	Full trade + data	P1 (M4)
Tradier	Native — REST (OAuth)	Full trade + data	P2
Fidelity / E*TRADE / Robinhood	Aggregator	Per broker	P2 (M5–6)
Additional brokers	Aggregator	Per broker	Post-launch

APPROVALS

Native programs (Schwab, IBKR) begin approval in P0/M1 — a long-lead item started early. The aggregator delivers breadth in parallel so coverage is never gated on one approval.

Billing & entitlements platform.

Billing engine

- Stripe-class processor
- Monthly & annual cycles
- Proration & upgrades
- Trials & coupons

Entitlements

- Central feature flags
- Per-tier gating
- Usage limits
- Real-time enforcement

Lifecycle

- Self-serve signup
- Dunning & recovery
- Cancellation & win-back
- Invoices & receipts

DEPENDENCY

Entitlements ride the per-tenant flag backbone built in P1 (M3–4); billing plugs into it in P2 (M5). This is why subscriptions can go live cleanly the moment onboarding is ready.

Mobile apps & admin console.

Mobile apps (iOS & Android)

Native-shell with a shared design system — full trading, alerts, push notifications, biometric login, and glanceable home-screen widgets. Store-ready by end of P3.

Admin & support console

Tenant management, entitlement overrides, health dashboards, immutable audit trails, and secure support workflows with OTP-protected sensitive actions.

FOUNDATION REUSED

The current platform already ships a responsive web app (light/dark), an Android app, and an OTP-protected admin flow. P3 extends these to full native mobile and a multi-tenant admin console.

Testing & QA — defense in depth.

Quality is engineered in layers, each catching what the layer below cannot — culminating in real traders on real brokers under observation.

ALL PHASES

Automated suites

Unit, integration and end-to-end tests gate every release; order & P&L logic carries the deepest coverage.

P1-P2

Broker sandbox conformance

Every adapter validated against the broker's sandbox (auth, quotes, order lifecycle, edge cases) before touching live money.

P3 · P5

Load, soak & chaos

High-concurrency load, multi-day soak, and failure injection (broker outages, session drops) to prove graceful degradation.

P4

Real-money closed beta

Instrumented real traders on real brokers — the failure modes a sandbox can't surface.

P5

Security & DR

Third-party penetration test; disaster-recovery and failover drills.

EXIT CRITERIA, NOT A CALENDAR

Each testing phase advances only when explicit, measurable bars are met — order-success, session-uptime, data-accuracy, zero critical security findings.

Signup to first trade, under five minutes.

The self-serve flow built in P2 is engineered to remove every point of friction between "sign up" and "place a live order."

STEP 1

Sign up & verify

Create tenant, verify email, secure the account.

STEP 2

Choose a plan

Start a trial or subscribe; entitlements set instantly.

STEP 3

Connect broker (OAuth)

Authorize on the broker's own page; a scoped, revocable token lands in the vault — no password stored.

STEP 4

Sync account

Positions, balances and buying power load; the terminal populates.

STEP 5

Trade

Alerts and the live order desk are active — the trader is fully operational.

WHY THIS MATTERS TO THE TIMELINE

Onboarding completion rate is a P2/P4 KPI. A frictionless flow is what turns the built platform into a business that a stranger can adopt unaided.

The whole program, one row per month.

MONTH	PHASE	PRIMARY FOCUS THAT MONTH
M1	P0	Architecture, adapter spec, security model, CI/CD; broker approvals started
M2	P1	Multi-tenancy foundation live; IBKR engine behind tenant boundary
M3	P1	Broker abstraction implemented; IBKR re-platformed onto it
M4	P1→P2	Schwab + Alpaca adapters; session pool; onboarding begins
M5	P2	Billing & entitlements; aggregator brokers online
M6	P2→P3	Statements/notifications; shared market-data feed begins
M7	P3	Scale infra + observability; AI co-pilot; risk guardrails; mobile
M8	P3	Hardening, load tests, pen-test prep — build complete
M9	P4	Closed beta onboarding — real traders, real brokers
M10	P4	Feedback loop & rapid iteration
M11	P4	Reliability & support playbooks; widen cohort
M12	P4	Exit-criteria review
M13	P5	Expanded beta; load / soak / chaos testing
M14	P5	Third-party pen-test; DR & failover drills
M15	P5	Compliance / SOC 2 readiness; final sign-off — launch-ready

How the timeline is meant to be read.

ASSUMPTIONS & BUFFER

- Month-based dates are **targets carrying a built-in comfort buffer of up to ~3 months** — the program is committed to land within the buffered window.
- The proven core (live brokerage engine) is reused, not rebuilt — the largest schedule risk is already retired.
- Native broker availability depends on third-party API approvals (started M1); the aggregator path de-risks this.
- Market-data vendor terms are validated in P3 before the shared feed goes live.
- Every phase closes on a demonstrable checkpoint; a phase advances only when its exit criteria are met.

GLOSSARY

Tenant

An isolated customer account on the shared platform.

Broker Adapter

The common interface every broker integration implements.

Aggregator

A regulated service (SnapTrade / Plaid-class) connecting many brokers through one integration.

OAuth

Password-less authorization — the trader consents on the broker's own login; NOVA gets a scoped, revocable token.

RLS

Row-Level Security — the database enforces tenant boundaries per row.

MIT

Market-If-Touched — when price touches a level, a market order fires.

SLO

Service-Level Objective — a reliability target the platform commits to.

Checkpoint

A demonstrable deliverable that closes a phase.

A proven engine, scaled on a schedule you can hold.

Fifteen months, six phases, seven parallel workstreams, and more than twenty demonstrable milestones — every one mapped to the point in time it happens.

THE BUILD — MONTHS 1-8

- P0 foundation & architecture
- P1 multi-tenancy + broker abstraction
- P2 subscriptions + onboarding + breadth
- P3 scale, data, AI, mobile & hardening

THE PROVING — MONTHS 9-15

- P4 closed beta with real traders on real brokers
- P5 scale, security, DR & compliance
- Staged go-live once SLOs hold
- Funded post-launch roadmap thereafter

— WHY THE SCHEDULE IS CREDIBLE

The hard part is already built.

A live, reliable brokerage engine — the thing that sinks most trading-app timelines — already runs in production on real money. This program scales proven technology on a milestone-gated schedule, with a comfort buffer built in, and closes on a launch-ready, multi-tenant, multi-broker platform.

15_{mo}

END TO END

6

PHASES

20₊

MILESTONES

M15

LAUNCH-READY